

Theorem (Cauchy–Schwarz Inequality). For real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

with equality iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i .

Proof. 1. **Interpretation as vectors and inner product.** Let

$$a = (a_1, \dots, a_n), \quad b = (b_1, \dots, b_n) \in \mathbb{R}^n.$$

On $\mathbb{R}^n$ consider the standard inner product

$$\langle a, b \rangle := \sum_{i=1}^n a_i b_i,$$

and the induced Euclidean norm

$$\|a\| := \sqrt{\langle a, a \rangle} = \sqrt{\sum_{i=1}^n a_i^2}, \quad \|b\| := \sqrt{\sum_{i=1}^n b_i^2}.$$

2. **Non-negativity of a squared norm.** For any real scalar t the vector $a - tb$ satisfies $\|a - tb\|^2 \geq 0$. Expanding using the inner product gives

$$\begin{aligned} \|a - tb\|^2 &= \langle a - tb, a - tb \rangle \\ &= \langle a, a \rangle - 2t \langle a, b \rangle + t^2 \langle b, b \rangle \\ &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2. \end{aligned}$$

3. **Quadratic in t .** Define

$$Q(t) := \left(\sum_{i=1}^n b_i^2\right) t^2 - 2 \left(\sum_{i=1}^n a_i b_i\right) t + \left(\sum_{i=1}^n a_i^2\right).$$

From the previous step $Q(t) = \|a - tb\|^2 \geq 0$ for every real t . If $\sum b_i^2 = 0$ then $b = 0$ and the inequality is trivial, so we may assume the leading coefficient is positive.

4. **Discriminant condition.** A quadratic $\alpha t^2 + \beta t + \gamma$ with $\alpha > 0$ that is non-negative for all $t \in \mathbb{R}$ must satisfy $\beta^2 - 4\alpha\gamma \leq 0$. Here

$$\alpha = \sum b_i^2, \quad \beta = -2 \sum a_i b_i, \quad \gamma = \sum a_i^2.$$

Hence

$$\begin{aligned} \Delta &= \beta^2 - 4\alpha\gamma \\ &= (-2 \sum a_i b_i)^2 - 4 \left(\sum b_i^2\right) \left(\sum a_i^2\right) \\ &= 4 \left[\left(\sum a_i b_i\right)^2 - \left(\sum a_i^2\right) \left(\sum b_i^2\right) \right]. \end{aligned}$$

Since $\Delta \leq 0$, we obtain

$$\left(\sum a_i b_i\right)^2 \leq \left(\sum a_i^2\right)\left(\sum b_i^2\right),$$

which is precisely the Cauchy–Schwarz inequality.

5. **Equality case.** Equality holds iff $\Delta = 0$, i.e. iff the minimizing

$$t_0 = \frac{\sum a_i b_i}{\sum b_i^2}$$

satisfies $\|a - t_0 b\|^2 = 0$. This is equivalent to $a = t_0 b$, or $a_i = t_0 b_i$ for every i . Conversely, if $a_i = \lambda b_i$ for some $\lambda \in \mathbb{R}$, then both sides of the inequality coincide. Hence equality occurs exactly when the vectors a and b are linearly dependent.

□